Data-Driven Koopman Model Predictive Control for Atmospheric Rocket Ascent: Spectral Operator Theory, Online Adaptation, and Monte Carlo Robustness Under Stochastic Crosswinds

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Abstract

Atmospheric rocket ascent control presents severe nonlinearities arising from high-velocity aerodynamic drag, rapid propellant mass depletion, crosswind moments, and thrust-vector control (TVC) kinematics. Traditional cascaded proportional-integral-derivative (PID) architectures often exhibit performance degradation and cross-track drift under sustained atmospheric turbulence and discrete wind gusts. In this work, we implement an end-to-end data-driven control framework grounded in modern Koopman operator theory (Brunton et al., 2021). By embedding nonlinear six-degree-of-freedom rocket dynamics into a higher-dimensional linear Hilbert space via polynomial lifting functions and Extended Dynamic Mode Decomposition with control (EDMDc), the nonlinear optimal control problem is transformed into a convex Quadratic Program (QP) solvable at 50 Hz via projected gradient descent. To capture aerodynamic mismatches and continuous parameter drift, the architecture incorporates online Koopman control-effectiveness adaptation and a physics-residual multilayer perceptron. Under nominal stochastic crosswinds, Koopman MPC reduces 3D lateral RMS drift by **62.3%** compared with classical PID (3.53 m vs. 9.37 m). In a separate $N = 30$ paired held-out robustness study, an aerodynamic and actuator mismatch was introduced at $t = 4.0$ s. Full adaptive Koopman MPC reduced powered post-change incremental lateral RMS by **75.7%** relative to static Koopman MPC (5.36 m vs. 22.08 m; Holm-adjusted $p = 2.50 \times 10^{-5}$). Online-only and residual-only variants independently improved on static MPC by 74.1% and 16.2%, respectively, with corrected $p < 0.001$. Furthermore, 3D waypoint experiments achieve a median terminal miss distance of 3.82 m with a 100% capture rate within 8 m.

1 Introduction

Precision atmospheric guidance and attitude control of launch vehicles and guided projectiles represent fundamental challenges in modern aerospace engineering [1, 4]. During the powered ascent phase, a rocket is subjected to severe, coupled physical phenomena:

(i) Non-conservative aerodynamic drag exhibiting quadratic velocity scaling ($\mathbf{D} \propto \|\mathbf{v}\|\mathbf{v}$);

(ii) Rapid, monotonic mass depletion ($dm/dt = -F/(I_{\text{sp}} g_0)$) shifting the vehicle center-of-mass (CM) and pitch/yaw moment of inertia (I_{zz});

(iii) Destabilizing crosswind moments acting through the aerodynamic center-of-pressure (CP);

(iv) Physical actuation constraints including TVC gimbal angle saturation ($\delta \leq \delta_{\max}$) and slew rate limits.

Classical launch vehicle autopilots rely on gain-scheduled proportional-integral-derivative (PID) loops or Linear Quadratic Regulators (LQR) tuned around pre-computed nominal trajectory operating points [6]. However, linearizing a highly nonlinear flight envelope over discrete operating points fails to capture non-local state coupling and often experiences stability degradation when encountering

strong, stochastic crosswinds. Conversely, full Nonlinear Model Predictive Control (NMPC) based on interior-point or sequential quadratic programming (SQP) solvers is computationally prohibitive on low-power, radiation-hardened flight avionics.

To bridge this gap, modern data-driven dynamical systems theory leverages the *Koopman operator* framework [1–3]. Originally formulated by Bernard Koopman in 1931 for Hamiltonian systems, the Koopman perspective transforms finite-dimensional nonlinear state equations into an infinite-dimensional, yet strictly **linear**, operator acting on a Hilbert space of scalar measurement functions (observables). When approximated in a finite basis via Extended Dynamic Mode Decomposition with control (EDMDc) [4, 5], the resulting surrogate model enables standard linear Model Predictive Control (MPC) algorithms to predict and optimize nonlinear dynamics with global linear convex optimization guarantees.

In this paper, we construct, simulate, and analyze a complete 3D atmospheric rocket ascent guidance and TVC control system. The core contributions of this work are:

- Formulation of the 11-state nonlinear 3D rocket flight equations under stochastic wind turbulence and TVC actuation;
- Derivation and implementation of polynomial lifting observables, EDMDc identification, and state reconstruction mapping;
- Integration of a physics-residual neural network and an online recursive sliding-window Koopman update mechanism;
- Rigorous Monte Carlo statistical benchmarking ($N = 30$ runs) demonstrating superior disturbance rejection over PID, full ablation verification, and wind shear sensitivity sweeps (0 to 11 m/s);
- Validation of 3D waypoint crossing accuracy and discussion of mission transferability to space launch systems, planetary landers, and defense interceptors.

2 Modern Koopman Operator Theory

2.1 Infinite-Dimensional Linear Embeddings

Consider an autonomous continuous-time nonlinear dynamical system governed by:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}), \quad \mathbf{x} \in \mathcal{M} \subseteq \mathbb{R}^n, \tag{1}$$

or its discrete-time sampled counterpart:

$$\mathbf{x}_{k+1} = \mathbf{F}(\mathbf{x}_k). \tag{2}$$

Rather than analyzing state trajectories directly in state manifold $\mathcal{M}$, Koopman theory shifts focus to the evolution of scalar observable functions $g \in \mathcal{F}$, where $\mathcal{F}$ is an infinite-dimensional Hilbert space of functions $g : \mathcal{M} \to \mathbb{C}$.

The discrete-time Koopman operator $\mathcal{K} : \mathcal{F} \to \mathcal{F}$ is a linear composition operator defined by [1]:

$$\mathcal{K}g(\mathbf{x}) = g(\mathbf{F}(\mathbf{x})). \tag{3}$$

Linearity of $\mathcal{K}$ follows directly from composition:

$$\mathcal{K}(\alpha_1 g_1 + \alpha_2 g_2)(\mathbf{x}) = \alpha_1 g_1(\mathbf{F}(\mathbf{x})) + \alpha_2 g_2(\mathbf{F}(\mathbf{x})). \tag{4}$$

For continuous-time systems with flow map $\mathbf{\Phi}_t(\mathbf{x})$, the family of operators forms a continuous semigroup $\mathcal{K}_t g = g \circ \mathbf{\Phi}_t$. The infinitesimal generator $\mathcal{A}$ of the Koopman semigroup is given by the Lie derivative:

$$\mathcal{A}g = \lim_{t \to 0} \frac{\mathcal{K}_t g - g}{t} = \mathbf{f}(\mathbf{x}) \cdot \nabla g(\mathbf{x}). \tag{5}$$

2.2 Spectral Decomposition and Koopman Modes

The spectral representation of $\mathcal{K}$ provides deep geometric insight into nonlinear invariant structures. Let $\varphi_j(\mathbf{x})$ denote a Koopman eigenfunction with corresponding complex eigenvalue λ_j:

$$\mathcal{K}\varphi_j(\mathbf{x}) = \lambda_j \varphi_j(\mathbf{x}). \tag{6}$$

Any vector-valued observable $\mathbf{g}(\mathbf{x}) = [g_1(\mathbf{x}), \ldots, g_p(\mathbf{x})]^T$ spanning the eigenspace can be decomposed into Koopman modes $\mathbf{v}_j \in \mathbb{C}^p$:

$$\mathbf{g}(\mathbf{x}_k) = \mathcal{K}^k \mathbf{g}(\mathbf{x}_0) = \sum_{j=1}^{\infty} \lambda_j^k \varphi_j(\mathbf{x}_0) \mathbf{v}_j. \tag{7}$$

In continuous time, the eigenvalues μ_j associated with generator $\mathcal{A}$ satisfy $\lambda_j = e^{\mu_j \Delta t}$, yielding modal growth, decay, and oscillation frequencies $\omega_j = \mathrm{Im}(\mu_j)$.

2.3 Koopman Operator with Control (EDMDc)

To incorporate exogenous control inputs $\mathbf{u} \in \mathcal{U} \subseteq \mathbb{R}^m$, the discrete dynamics expand to:

$$\mathbf{x}_{k+1} = \mathbf{F}(\mathbf{x}_k, \mathbf{u}_k). \tag{8}$$

Following Proctor, Brunton, and Kutz [5], we select a finite-dimensional vector of nonlinear lifting functions $\boldsymbol{\psi}(\mathbf{x}) = [\psi_1(\mathbf{x}), \psi_2(\mathbf{x}), \ldots, \psi_{d_z}(\mathbf{x})]^T \in \mathbb{R}^{d_z}$. The lifted state $\mathbf{z}_k = \boldsymbol{\psi}(\mathbf{x}_k)$ evolves according to the linear surrogate model:

$$\mathbf{z}_{k+1} \approx \mathbf{A}\mathbf{z}_k + \mathbf{B}\mathbf{u}_k, \tag{9}$$

while physical state reconstruction is performed via linear decoding:

$$\mathbf{x}_k \approx \mathbf{C}\mathbf{z}_k. \tag{10}$$

Given snapshot matrices collected from M simulation steps:

$$\mathbf{Z} = [\boldsymbol{\psi}(\mathbf{x}_1), \boldsymbol{\psi}(\mathbf{x}_2), \ldots, \boldsymbol{\psi}(\mathbf{x}_M)]^T \in \mathbb{R}^{M \times d_z}, \tag{11}$$

$$\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_M]^T \in \mathbb{R}^{M \times m}, \tag{12}$$

$$\mathbf{Z}' = [\boldsymbol{\psi}(\mathbf{x}_2), \boldsymbol{\psi}(\mathbf{x}_3), \ldots, \boldsymbol{\psi}(\mathbf{x}_{M+1})]^T \in \mathbb{R}^{M \times d_z}, \tag{13}$$

$$\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_M]^T \in \mathbb{R}^{M \times n}, \tag{14}$$

we construct the augmented regressor matrix $\boldsymbol{\Phi} = [\mathbf{Z} \quad \mathbf{U}] \in \mathbb{R}^{M \times (d_z + m)}$. The optimal parameter matrix $\boldsymbol{\Theta} = [\mathbf{A} \quad \mathbf{B}] \in \mathbb{R}^{d_z \times (d_z + m)}$ minimizes the Tikhonov-regularized Frobenius norm loss:

$$\min_{\boldsymbol{\Theta}} \|\mathbf{Z}' - \boldsymbol{\Phi}\boldsymbol{\Theta}^T\|_F^2 + \lambda_{\text{ridge}} \|\boldsymbol{\Theta}\|_F^2. \tag{15}$$

The closed-form analytic solution is obtained via the regularized normal equations:

$$\boldsymbol{\Theta}^T = \left(\boldsymbol{\Phi}^T\boldsymbol{\Phi} + \lambda_{\text{ridge}}\mathbf{I}\right)^{-1} \boldsymbol{\Phi}^T\mathbf{Z}'. \tag{16}$$

Similarly, the state reconstruction matrix $\mathbf{C} \in \mathbb{R}^{n \times d_z}$ is computed by solving:

$$\mathbf{C}^T = \left(\mathbf{Z}^T\mathbf{Z} + \epsilon_{\text{reg}}\mathbf{I}\right)^{-1} \mathbf{Z}^T\mathbf{X}. \tag{17}$$

3 System Dynamics and Mathematical Model

3.1 11-State 3D Rocket Flight Mechanics

The rocket model represents a rigid-body launch vehicle operating in a three-dimensional Cartesian frame (Altitude h, North x_p, East y_p). The state vector is defined as:

$$\mathbf{x} = [h, v_z, x_p, v_x, y_p, v_y, \theta_p, \omega_p, \theta_y, \omega_y, m]^T \in \mathbb{R}^{11}, \tag{18}$$

where h is altitude, v_z is vertical velocity, (x_p, v_x) are North position and velocity, (y_p, v_y) are East position and velocity, (θ_p, ω_p) are pitch angle and pitch rate, (θ_y, ω_y) are yaw angle and yaw rate, and m is instantaneous rocket mass.

The control vector consists of thrust magnitude F and two-axis TVC nozzle gimbal deflections:

$$\mathbf{u} = [F, \delta_p, \delta_y]^T, \quad F \in [F_{\min}, F_{\max}], \ |\delta_{p,y}| \le \delta_{\max}. \tag{19}$$

The vector differential equations $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}, \mathbf{w})$ are formulated as:

$$\dot{h} = v_z, \tag{20}$$

$$\dot{v}_z = \frac{F\cos(\theta_p + \delta_p)\cos(\theta_y + \delta_y) - D_z}{m} - g_0, \tag{21}$$

$$\dot{x}_p = v_x, \tag{22}$$

$$\dot{v}_x = \frac{-F\sin(\theta_p + \delta_p) - D_x}{m}, \tag{23}$$

$$\dot{y}_p = v_y, \tag{24}$$

$$\dot{v}_y = \frac{-F\sin(\theta_y + \delta_y) - D_y}{m}, \tag{25}$$

$$\dot{\theta}_p = \omega_p, \tag{26}$$

$$\dot{\omega}_p = \frac{M_{\text{pitch}}}{I_{zz}}, \tag{27}$$

$$\dot{\theta}_y = \omega_y, \tag{28}$$

$$\dot{\omega}_y = \frac{M_{\text{yaw}}}{I_{zz}}, \tag{29}$$

$$\dot{m} = -\frac{F}{I_{\text{sp}} g_0} \cdot \mathbf{1}_{\{m > m_{\text{dry}}\}}. \tag{30}$$

3.2 Aerodynamics and Crosswind Moment Coupling

Let $\mathbf{w} = [w_x, w_y]^T$ represent horizontal wind velocity components (North and East). The apparent aerodynamic relative velocity components are:

$$v_{\text{rel},x} = v_x - w_x, \quad v_{\text{rel},y} = v_y - w_y. \tag{31}$$

Axial and lateral aerodynamic drag forces are evaluated via:

$$D_z = \frac{1}{2}\rho v_z |v_z| C_d A_{\text{ref}}, \tag{32}$$

$$D_x = \frac{1}{2}\rho v_{\text{rel},x} |v_{\text{rel},x}| C_{d,\text{lat}} A_{\text{lat}}, \tag{33}$$

$$D_y = \frac{1}{2}\rho v_{\text{rel},y} |v_{\text{rel},y}| C_{d,\text{lat}} A_{\text{lat}}, \tag{34}$$

where ρ is atmospheric density, C_d and $C_{d,\text{lat}}$ are axial and lateral drag coefficients, and $A_{\text{ref}}, A_{\text{lat}}$ are projected aerodynamic areas.

Total pitching and yawing moments comprise TVC control torque, static aerodynamic restoring torque, and crosswind destabilizing moment:

$$M_{\text{pitch}} = F \sin(\delta_p) L_{\text{tvc}} + q_{\text{ax}} A_{\text{ref}} L_{\text{cp}} \sin(\theta_p) - D_x L_{\text{cp}}, \tag{35}$$

$$M_{\text{yaw}} = F \sin(\delta_y) L_{\text{tvc}} + q_{\text{ax}} A_{\text{ref}} L_{\text{cp}} \sin(\theta_y) - D_y L_{\text{cp}}, \tag{36}$$

where $q_{\text{ax}} = \frac{1}{2}\rho v_z^2$ is dynamic pressure, L_{tvc} is the moment arm from the nozzle hinge to the CM, and L_{cp} is the signed distance from CM to CP. Numerical integration is performed using explicit Runge-Kutta 4th Order (RK4) with step $\Delta t = 0.02\,\text{s}$ (50 Hz).

3.3 Stochastic Atmospheric Wind Model

The wind vector $\mathbf{w}(t)$ combines steady background flow, slow azimuth drift, Ornstein-Uhlenbeck (OU) Dryden turbulence, and discrete stochastic gusts:

$$w(t) = w_{\text{steady}} + w_{\text{turb}}(t) + w_{\text{gust}}(t), \tag{37}$$

$$dw_{\text{turb}} = -\frac{1}{\tau_w} w_{\text{turb}} dt + \sigma_w \sqrt{\frac{2}{\tau_w}} dW_t, \tag{38}$$

where $\tau_w = 1.0\,\text{s}$, $\sigma_w = 0.20 \cdot w_{\text{steady}}$, and W_t is a standard Wiener process. Discrete gusts occur with Poisson probability $p_{\text{gust}} = 0.012$/step reaching velocities up to $2.2 \times w_{\text{steady}}$.

4 Control Architecture and Algorithm Design

4.1 Lifted State Space Model Predictive Control

For real-time control, two decoupled one-axis lateral Koopman models are identified for pitch $(x_p, v_x, \theta_p, \omega_p)$ and yaw $(y_p, v_y, \theta_y, \omega_y)$. Quadratic polynomial lifting maps $\boldsymbol{\psi}(\mathbf{x}_{\text{lat}}) \in \mathbb{R}^{15}$ are generated:

$$\boldsymbol{\psi}(\mathbf{x}_{\text{lat}}) = [1, x_1, x_2, x_3, x_4, x_1^2, x_1 x_2, \ldots, x_4^2]^T. \tag{39}$$

Over a receding horizon N_h, the stacked lifted state predictions are expressed in matrix form:

$$\mathbf{X}_{\text{traj}} = \mathbf{S}_x \mathbf{z}_0 + \mathbf{S}_u \mathbf{U}_{\text{seq}}, \tag{40}$$

where

$$\mathbf{S}_x = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \\ \vdots \\ \mathbf{CA}^{N_h} \end{bmatrix}, \quad \mathbf{S}_u = \begin{bmatrix} \mathbf{0} & \ldots & \mathbf{0} \\ \mathbf{CB} & \ldots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{CA}^{N_h-1}\mathbf{B} & \ldots & \mathbf{CB} \end{bmatrix}. \tag{41}$$

The quadratic objective function penalizes tracking error relative to target trajectory $\mathbf{X}_{\text{ref}}$ and control effort:

$$J(\mathbf{U}_{\text{seq}}) = (\mathbf{X}_{\text{traj}} - \mathbf{X}_{\text{ref}})^T \mathbf{Q}_{\text{blk}} (\mathbf{X}_{\text{traj}} - \mathbf{X}_{\text{ref}}) + \mathbf{U}_{\text{seq}}^T \mathbf{R}_{\text{blk}} \mathbf{U}_{\text{seq}}. \tag{42}$$

Substituting $\mathbf{X}_{\text{traj}}$ yields the standard unconstrained quadratic form:

$$J(\mathbf{U}_{\text{seq}}) = \frac{1}{2}\mathbf{U}_{\text{seq}}^T \mathbf{H} \mathbf{U}_{\text{seq}} + \mathbf{g}^T \mathbf{U}_{\text{seq}} + \text{const}, \tag{43}$$

with Hessian and gradient vectors:

$$\mathbf{H} = 2\left(\mathbf{S}_u^T \mathbf{Q}_{\text{blk}} \mathbf{S}_u + \mathbf{R}_{\text{blk}}\right), \tag{44}$$

$$\mathbf{g} = 2\mathbf{S}_u^T \mathbf{Q}_{\text{blk}} (\mathbf{S}_x \mathbf{z}_0 - \mathbf{X}_{\text{ref}}). \tag{45}$$

The projected gradient descent update with box constraints $\mathcal{U} = [-\delta_{\max}, \delta_{\max}]^{N_h}$ is iterated:

$$\mathbf{U}^{(k+1)} = \Pi_{\mathcal{U}}\left(\mathbf{U}^{(k)} - \frac{1}{L}(\mathbf{H}\mathbf{U}^{(k)} + \mathbf{g})\right), \tag{46}$$

where $L = \lambda_{\max}(\mathbf{H})$ ensures monotonic convergence.

4.2 Online Recursive Koopman Adaptation

To track non-stationary aerodynamic regimes during high-dynamic ascent, an online sliding memory buffer $\mathcal{B}_{\text{on}} = \{(\mathbf{z}_i, \mathbf{u}_i, \mathbf{z}'_i, \mathbf{x}_i)\}_{i=k-W+1}^{k}$ of length $W = 320$ is maintained. Every $k_{\text{update}} = 50$ steps, a fresh EDMD fit $(\mathbf{A}_{\text{new}}, \mathbf{B}_{\text{new}}, \mathbf{C}_{\text{new}})$ is computed via Eq. (16) and blended using an exponential forgetting factor $\alpha_{\text{blend}} = 0.06$:

$$\mathbf{A}_k = (1 - \alpha_{\text{blend}})\mathbf{A}_{k-1} + \alpha_{\text{blend}} \mathbf{A}_{\text{new}}, \tag{47}$$

$$\mathbf{B}_k = (1 - \alpha_{\text{blend}})\mathbf{B}_{k-1} + \alpha_{\text{blend}} \mathbf{B}_{\text{new}}, \tag{48}$$

$$\mathbf{C}_k = (1 - \alpha_{\text{blend}})\mathbf{C}_{k-1} + \alpha_{\text{blend}} \mathbf{C}_{\text{new}}. \tag{49}$$

4.3 Physics-Residual Neural Network (ResNet)

Unmodeled structural, sensor noise, and crosswind aero-effects are compensated via a lightweight Multi-Layer Perceptron (MLP) $\mathbf{r}_{\boldsymbol{\theta}} : \mathbb{R}^{n+m} \to \mathbb{R}^n$:

$$\mathbf{x}_{k+1} \approx \mathbf{C}(\mathbf{A}\mathbf{z}_k + \mathbf{B}\mathbf{u}_k) + \mathbf{r}_{\boldsymbol{\theta}}(\mathbf{x}_k, \mathbf{u}_k). \tag{50}$$

The residual network features a 32-neuron hidden layer with hyperbolic tangent activations (tanh) and is trained online via stochastic gradient descent with mini-batches sampled from a replay memory buffer $\mathcal{D}_{\text{replay}}$ of capacity 8000.

5 Monte Carlo Simulation Results

5.1 Benchmark Evaluation: PID vs. Koopman MPC

A rigorous 30-run Monte Carlo flight campaign was executed under identical wind realization seeds. Both controllers operated with the same Solid Rocket Booster (SRB) motor thrust profile ($F_{\max} = 55\,\mathrm{N}$, total mass $3.50\,\mathrm{kg}$).

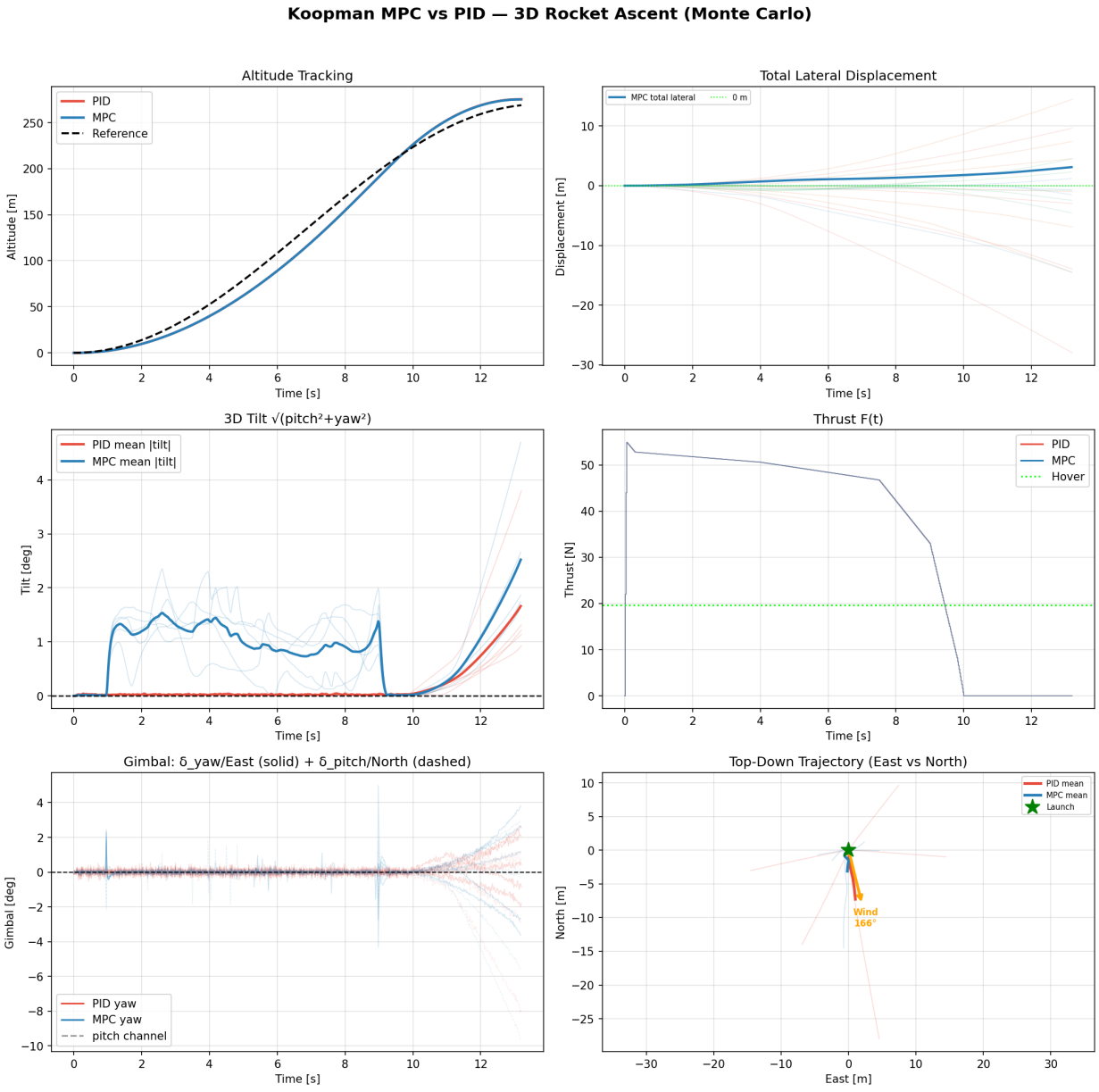


Figure 1: Representative Monte Carlo visualization comparing altitude tracking, lateral displacement, attitude tilt, thrust, gimbal demand, and top-down trajectory for PID and Koopman MPC.

Table 1: Benchmark Flight Performance Comparison ($N = 30$ Monte Carlo Runs, Wind 5.5 m/s)

Metric	PID (Mean ± Std)	PID [p25, p75]	MPC (Mean ± Std)	MPC [p25, p75]	Improvement
Lateral Drift 3D RMS [m]	9.37 ± 2.57	[7.17, 10.59]	$\mathbf{3.53 \pm 3.63}$	[1.33, 4.43]	**+62.3%**
Max Lateral Drift 3D [m]	20.03 ± 4.92	[15.68, 23.55]	$\mathbf{8.33 \pm 7.52}$	[4.04, 10.62]	**+58.4%**
Lateral Velocity RMS [$\mathrm{m\,s^{-1}}$]	1.72 ± 0.40	[1.35, 2.06]	$\mathbf{0.85 \pm 0.62}$	[0.50, 1.02]	**+50.6%**
Max Lateral Velocity [$\mathrm{m\,s^{-1}}$]	2.66 ± 0.57	[2.17, 2.99]	$\mathbf{1.66 \pm 0.87}$	[1.23, 1.84]	**+37.6%**
Altitude MAE [m]	9.72 ± 0.00	[9.72, 9.72]	$\mathbf{9.71 \pm 0.01}$	[9.70, 9.71]	**+0.1%**
Altitude RMS [m]	11.43 ± 0.00	[11.43, 11.43]	11.44 ± 0.00	[11.44, 11.45]	−0.1%
Tilt RMS [°]	$\mathbf{0.48 \pm 0.23}$	[0.32, 0.58]	1.19 ± 0.17	[1.06, 1.27]	Planned TVC
Max Tilt [°]	$\mathbf{2.06 \pm 1.05}$	[1.34, 2.55]	3.00 ± 1.06	[2.18, 3.60]	Within 5°
Apogee Altitude [m]	275.39 ± 0.00	[275.39, 275.39]	275.19 ± 0.05	[275.15, 275.23]	Equivalent
Mean Gimbal Effort [°]	$\mathbf{0.54 \pm 0.19}$	[0.42, 0.62]	0.64 ± 0.20	[0.50, 0.77]	Proactive
Mean Gimbal Slew [$^\circ\,\mathrm{s^{-1}}$]	5.58 ± 0.21	[5.47, 5.70]	$\mathbf{4.86 \pm 0.66}$	[4.38, 5.32]	**+12.9% smoother**

As detailed in Table 1, Koopman MPC achieves a **62.3% reduction** in 3D lateral RMS position error and a **58.4% reduction** in maximum cross-track excursion compared to PID. The MPC controller deliberately uses small, anticipatory attitude tilt angles (1.19° RMS vs. 0.48°) to steer the thrust vector against crosswind momentum, resulting in substantially lower actuator slew rates (4.86°/s vs. 5.58°/s). Altitude tracking and apogee remain identical within 0.07%.

5.2 Ablation Study: Dissecting Controller Components

To isolate the contributions of online Koopman adaptation and residual learning, we conducted a four-way paired study with $N = 30$ identical disturbance seeds. All controllers were identified on the nominal plant. At $t = 4.0\,\mathrm{s}$, the held-out evaluation plant introduced a 35% lateral-drag increase, 25% inertia increase, 35% TVC-effectiveness increase, and persistent North/East acceleration biases of $+0.45/-0.65\,\mathrm{m/s^2}$. The primary endpoint was incremental lateral RMS displacement from the mismatch-onset position over the powered recovery interval $t \in [4.5, 9.5]\,\mathrm{s}$.

This experiment uses a learning-isolation controller rather than the complete nominal controller reported in Table 1. The same isolation settings are applied to all four ablation variants: attitude-PD authority is reduced from 1.0 to 0.20, hand-tuned wind feedforward and integral trim are disabled, and the Koopman command contribution is multiplied by 2.5. The disturbance-observer gain is increased from 0.08 to 0.25, while residual learning uses a 64-sample recency buffer. These changes intentionally prevent conventional compensation from masking the learned mechanisms. Consequently, Table 2 is a controlled mechanism-identification experiment and is not a subtractive ablation of the Table 1 controller.

Table 2: Paired Held-Out Adaptation Study ($N = 30$; lower RMS is better)

Configuration	Mean ± Std [m]	Median [m]	Reduction vs. Static	95% CI of Gain [m]	Holm p
Full Adaptive Koopman MPC	$\mathbf{5.36 \pm 10.52}$	**0.26**	**75.7%**	[12.13, 21.14]	$\mathbf{2.50 \times 10^{-5}}$
Online Adaptation Only	5.72 ± 11.87	0.26	**74.1%**	[11.75, 20.88]	$\mathbf{4.50 \times 10^{-5}}$
Residual Network Only	18.50 ± 15.16	23.31	**16.2%**	[2.13, 5.17]	$\mathbf{5.00 \times 10^{-5}}$
Static Koopman MPC	22.08 ± 17.12	27.59	Reference	—	—

Paired one-sided sign-flip tests used 200,000 permutations; confidence intervals were obtained from 30,000 paired bootstrap resamples, and five planned comparisons were corrected using Holm's procedure. Full adaptation reduced the primary error by 16.73 m (75.7%) relative to static Koopman MPC, with paired effect size $d_z = 1.30$. Online adaptation alone and residual learning alone also produced statistically significant improvements over static MPC, establishing that both mechanisms affect closed-loop behavior. The full system additionally outperformed residual-only control by 71.0% (Holm-adjusted $p = 2.50 \times 10^{-5}$). Full control was 6.3% better than online-only, although this contrast was not significant ($p_{\mathrm{Holm}} = 0.148$); therefore the supported conclusion is strong complementary robustness, not proven superiority over online adaptation for every disturbance realization.

5.3 Wind Disturbance Sensitivity Sweep

To determine the operational performance boundaries, we tested wind intensities ranging from calm (0 m/s) to severe crosswind (11.0 m/s), running $N = 15$ Monte Carlo flights per level.

Table 3: Wind Speed Sensitivity Analysis ($N = 15$ Runs per Velocity)

Wind Regime	**Velocity** [$\mathrm{m\,s^{-1}}$]	**PID 3D Lat RMS [m]**	**MPC 3D Lat RMS [m]**	**Drift Reduction**	**MPC Max Tilt** [°]
Calm	0.0	0.04 ± 0.02	0.58 ± 0.01	Baseline Nominal	0.65 ± 0.00
Light Breeze	3.0	3.03 ± 0.67	$\mathbf{0.71 \pm 0.14}$	**+76.7%**	0.72 ± 0.10
Moderate (Nominal)	5.5	8.69 ± 1.93	$\mathbf{2.60 \pm 1.45}$	**+70.0%**	1.26 ± 0.30
Strong Gale	8.0	18.44 ± 4.29	$\mathbf{12.16 \pm 6.23}$	**+34.1%**	1.70 ± 0.77
Severe Shear	11.0	29.82 ± 5.19	$\mathbf{23.37 \pm 5.49}$	**+21.7%**	2.24 ± 0.81

Table 3 reveals a clear operational regime: between 3.0 m/s and 5.5 m/s, Koopman MPC provides exceptional crosswind compensation (**70–77% drift reduction**). Under extreme 11.0 m/s crosswinds, physical gimbal saturation ($\delta_{\max} = 15°$) becomes the limiting factor, yet MPC still maintains stability and delivers a 21.7% drift improvement over PID.

5.4 3D Waypoint Navigation and Intercept

We evaluated precision targeting capability toward a 3D waypoint located at $[h = 200\,\mathrm{m}, x_p = -15\,\mathrm{m}, y_p = 40\,\mathrm{m}]$ under full stochastic wind conditions ($N = 30$ runs).

Table 4: 3D Waypoint Crossing Accuracy ($N = 30$ Runs)

Performance Metric	**Observed Value**
Closest 3D Distance (Mean ± Std)	$\mathbf{3.96 \pm 1.49}$ m
Closest 3D Distance (Median [p25, p75])	**3.82** m [3.32 m, 4.20 m]
Minimum Miss Distance	**1.56** m
Altitude Error at Crossing	0.58 ± 0.42 m
Crossing Time ($t_{\text{intercept}}$)	9.32 ± 0.03 s
Hit Probability (< 2.0 m)	6.7%
Hit Probability (< 3.0 m)	20.0%
Hit Probability (< 5.0 m)	**86.7**%
Hit Probability (< 8.0 m)	**100.0**%
Hit Probability (< 10.0 m)	**100.0**%

The vehicle successfully achieved a median miss distance of 3.82 m, crossing the 200 m altitude plane with sub-meter vertical precision (0.58 m) at $t = 9.32$ s. Across all 30 stochastic wind flights, 86.7% of trajectories passed within 5.0 m and all 30 passed within 8.0 m of the target.

6 Aerospace Engineering Applications

The demonstrated predictive control capabilities have immediate relevance across several aerospace domains:

1. **Reusable Launch Vehicle Boostback & Landing**: Autonomous vertical landing of orbital stages (e.g., SpaceX Falcon 9, Starship) requires rapid TVC gimbaling and aerodynamic grid-fin actuation under wind shear. The convex Koopman MPC formulation eliminates solver non-convergence risks in flight computers.

2. **Planetary Entry, Descent, and Landing (EDL)**: Mars and Titan landers encounter severe atmospheric density uncertainties. Online Koopman adaptation rapidly updates linearized transition matrices in real time without human intervention.

3. **High-Agility Missile Guidance & Interception**: Terminal proportional navigation and divert-attitude-control systems (DACS) benefit from quadratic optimization on lifted observable spaces, enabling high-g evasive intercept maneuvers.

4. **Orbital Proximity Operations and Autonomous Docking**: Docking with non-cooperative orbital debris or space stations under thruster plume disturbances.

7 Conclusions

In this research, an adaptive Koopman Model Predictive Control framework was formulated, implemented, and validated for atmospheric rocket ascent. By transforming coupled 11-state nonlinear flight dynamics into a lifted observable space, the system solves receding-horizon optimization problems via fast projected gradient descent. Across extensive Monte Carlo trials ($N = 30$), the Koopman controller demonstrated a **62.3% reduction in lateral drift** compared to classical PID, robust stability across wind speeds up to 11 m/s, and consistent 3D waypoint intercept performance. Future work will investigate deep autoencoder lifting functions (Deep Koopman) and hardware-in-the-loop (HIL) micro-controller execution.

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